

C Programming - Test

Time: 30 Minutes

Total Questions: 15

All questions are compulsory.

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1. Write a structure definition in C to store information of a book with fields: title, author, and price.

Ans:

```
struct book {  
    char title [50];  
    char author [20];  
    int price;  
};
```

2. Given the structure below, write the correct syntax to access the marks field:

```
struct Student {  
    int roll;  
    float marks;  
} s1;
```

Ans: s1.marks;

3. What is the size of the following structure on a 64-bit system?

```
struct Data {  
    char a;  
    int b;  
};
```

Ans: 8 bytes (int 4 bytes, char a 1 byte but due to padding total is 8 bytes)

4. Write a union definition that can store either an int, a float, or a char.

```
union data {  
    int a;  
    float b;  
    char ch;  
};
```

5. What is the output of the following bitwise operation?

`printf("%d", 5 & 3);`

$$\begin{array}{r} 5 = 0000\ 0101 \\ 3 = 0000\ 0011 \\ \hline 1 = 0000\ 0001 \end{array}$$

Ans: 1

6. What is the result of `12 | 5` in C?

Ans: `12 = 0000 1100` `5 = 0000 0101`

$$12 | 5 = 13$$

$$\begin{array}{r} 0000\ 1100 \rightarrow 12 \\ 0000\ 0101 \rightarrow 5 \\ \hline 0000\ 1101 \rightarrow 13 \end{array}$$

7. Which storage class makes a variable retain its value between function calls?

Ans: static

8. Write a single C statement to toggle the 2nd bit of variable x.

Ans: `x = x ^ (1 << 2);`

9. What is a NULL and dangling pointer in c write example to show these ?

NULL pointer: If a pointer variable is assigned to NULL then it is called NULL pointer.

eg:- `int *ptr = NULL;`

Dangling pointer: If a pointer is referred to free variable in dynamic memory is called Dangling pointer.

eg:- `free(Node); int *ptr = Node;` // `*ptr` is dangling pointer.

10. Write a C expression to check if the 3rd bit (from LSB) of a variable x is set.

$$x = (x \gg 3) \& 1;$$

Ans: `int check = (x >> 3) & 1 ? 1 : 0;`

11. How many times will the following loop run?

```
for (int i = 0; i < 5; i++)  
{  
    printf("%d", i);  
}
```

Ans: 5 times.

12. Which loop is guaranteed to run at least once in C?

Ans: do while loop

13. Given `int arr[5] = {10, 20, 30, 40, 50};`,

what is the value of `*(arr + 2)`?

Ans: `*(arr + 2)` value is 30

14. What will be printed?

`int x = 5, y = 2;`

`printf("%d", x >> y);`

`5 >> 2` `00000101 >> 2`

Ans: 1

15. Explain the difference between call by value and call by reference in C, write example code.

→ Call by value:-

If the copy of a variable is sent as parameter/argument to a function, then the function is called "call by value".

eg:- `int add(int a, int b);`

`int main()`

{ `int a = 20;`

`int b = 30;`

`int c = add(a, b);` // call by value)

}

`int add(int a, int b);`

{

`return a+b;`

}

→ call by reference:-

→ If the address of a variable is sent as argument to a function, then the function is called "call by reference".

eg:-

```
int add (int *a, int *b) {
```

```
    return *a + *b;
```

```
}
```

```
int main()
```

```
{    int a = 2;
```

```
    int b = 3;
```

```
    int c = add (&a, &b); // call by reference
```

```
    return 0;
```

```
}
```